

A Morse-Theoretic Construction of the Atiyah–Hirzebruch Spectral Sequence

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Notation & Conventions

This page collects all notational conventions used in this thesis. It is to be **removed before submission**.

1. Manifolds, Maps, and the Differential

Symbol	Meaning & Convention
M, N	smooth manifolds (always assumed without boundary unless stated)
$T_p M, T_p^* M$	tangent / cotangent space at $p \in M$
$TM, T^* M$	tangent / cotangent bundle
df	exterior differential of $f: M \rightarrow \mathbb{R}$; a 1-form on M
$d_p f = (df)_p$	differential of f at the point p ; a linear map $T_p M \rightarrow \mathbb{R}$. Rule: write $d_p f$ (macro <code>\dat{p}f</code>) when evaluating at a specific point. Write df for the full 1-form / bundle map.
$d_p F$	for $F: M \rightarrow N$: the linear map $T_p M \rightarrow T_{F(p)} N$. Also written $(dF)_p$; never write $\frac{d}{dx} F$ for a map between manifolds.
$\frac{d}{dt}$	total derivative w.r.t. a <i>single</i> real variable t (roman d, macro <code>\diff</code>). Use for curves, flows evaluated on a trajectory.
$\frac{\partial}{\partial t}$	partial derivative; use only when <i>at least two independent</i> variables/parameters are present. Never write $\partial/\partial t$ for the time-derivative along a flow line.
$\nabla_v f$	covariant derivative of f in direction v (Levi-Civita or chosen connection)
$\text{grad}_g f$	gradient of f with respect to Riemannian metric g ; characterised by $g(\text{grad}_g f, -) = df$

2. Flows and Dynamical Systems

Symbol	Meaning & Convention
$\varphi_t(x)$	flow of a vector field at time t through initial point x . Convention: always write φ_t , not $\varphi(t, \cdot)$; think of $\varphi_t \in \text{Diff}(M)$ as a one-parameter family of diffeomorphisms.
$\varphi_t = e^{tX}$	flow of the vector field X ; $\varphi_0 = \text{id}$.
$\frac{d}{dt} \varphi_t(x)$	time-derivative of the flow line $t \mapsto \varphi_t(x)$; equals $X(\varphi_t(x))$. Use roman d, never ∂ .
$W^u(p), W^s(p)$	unstable / stable manifold of a critical (rest) point p (macros <code>\Wu{p}</code> , <code>\Ws{p}</code>)

Symbol	Meaning & Convention
$\mathcal{M}(p, q)$	moduli space of (unparametrised) flow lines from p to q (macro <code>\Mflow{p}{q}</code>)

3. Restrictions and Evaluations

Core rule: distinguish carefully between a *fiber*, a *restriction to a subspace*, and a *value/evaluation*.

Symbol	Meaning & Convention
E_p	fiber of a vector bundle $E \rightarrow X$ over $p \in X$. Preferred over $E _{\{p\}}$ for fibers; use macro <code>E_p</code> .
$E _A$	restriction of the bundle E to the subspace $A \subseteq X$; this is itself a bundle over A . Macro: <code>\restr{E}{A}</code> .
$\xi _A$	section $\xi \in \Gamma(E)$ restricted to A ; yields $\xi _A \in \Gamma(E _A)$.
$\xi(p)$ or ξ_p	value of the section ξ at the point p ; an element of E_p . Avoid $\xi _p$ for this; use $\xi(p)$.
$\omega_p = \omega _p$	value of a differential k -form ω at p ; an element of $\Lambda^k T_p^* M$. Acceptable to write $\omega _p$ only when the point-evaluation is being <i>emphasised</i> over the fiber.
Summary	E_p = fiber (bundle); $E _A$ = restricted bundle; $\xi(p)$ = value of section; ω_p = value of form.

5. Chain Complexes and Homology Theories

Three homology theories appear simultaneously; they are always decorated.

Theory	Chain complex	Boundary / Homology
Singular	$({}^S C_*(X), {}^S \partial)$	boundary ${}^S \partial$; homology $H^S H_*(X)$. Macros: <code>\ChS</code> , <code>\bS</code> , <code>\HS</code> .
CW	$({}^{CW} C_*(X), {}^{CW} \partial)$	boundary ${}^{CW} \partial$; homology $H^{CW} H_*(X)$. Macros: <code>\ChCW</code> , <code>\bCW</code> , <code>\HCW</code> .
Morse	$({}^M C_*(f), {}^M \partial)$	boundary ${}^M \partial$; homology ${}^M H_*(f)$. Macros: <code>\ChM</code> , <code>\bM</code> , <code>\HM</code> .

Connecting homomorphisms in long exact sequences: $: {}^M \delta, {}^{CW} \delta, {}^S \delta$ with the macros: `\connM`, `\connCW`, `\connS`.

Co-homology: For co-homology we use upper indices on the co-boundary maps: ${}^M d, {}^{CW} d, {}^S d$ with the macros: `\cbM`, `\cbCW`, `\cbS`. And for the co-homology we use the homologie macros with upper indices.

Theory	Chain complex	Boundary / Homology
Filtration: $F^{p\mathcal{S}}C_*(X)$ for the filtration of the singular complex; boundary is still ∂ .		

6. Vector Bundles and K-Theory

Symbol	Meaning
$\text{Vect}(X)$	set of isomorphism classes of complex vector bundles over X (macro <code>\Kvect{X}</code>)
$[E]$	stable isomorphism class / element in $K(X)$
$K(X) = KU(X)$	complex K-theory (default); macro <code>\K</code> , <code>\KU</code>
$KO(X)$	real K-theory; macro <code>\KO</code>
$\tilde{K}(X)$	reduced complex K-theory
$E \oplus F, E \otimes F$	Whitney sum / tensor product of bundles
$E _A$	restriction of bundle $E \rightarrow X$ to subspace A (same convention as §3)
E_p	fiber of E over p (same convention as §3)

7. Spectral Sequences

Symbol	Meaning
$E_{p,q}^r$	(p, q) -entry on the r -th page of a spectral sequence; macro <code>\Epq{r}{p,q}</code>
d^r	differential on the r -th page; $d^r: E_{p,q}^r \rightarrow E_{p-r, q+r-1}^r$ (cohomological convention); macro <code>\drpq{r}</code>
AHSS	Atiyah–Hirzebruch spectral sequence: $E_{p,q}^2 \cong H^p(X; K^q(\text{pt})) \Rightarrow K^{p+q}(X)$
$E_{p,q}^\infty$	abutment / E_∞ -page

8. Frequently Used Macros (Quick Reference)

Macro	Output	Use
<code>\diff</code>	d	exterior deriv. / total deriv.
<code>\dat{p}</code>	d_p	deriv. at point p
<code>\restr{E}{A}</code>	$E _A$	restriction
<code>\Wu{p}</code>	$W^u(p)$	unstable manifold

Macro	Output	Use
<code>\Ws{p}</code>	$W^s(p)$	stable manifold
<code>\Mflow{p}{q}</code>	$\mathcal{M}(p, q)$	moduli of flow lines
<code>\ChM</code>	$\mathcal{M}C_*$	Morse chain complex
<code>\bM</code>	$\mathcal{M}\partial$	Morse boundary
<code>\HM</code>	$\mathcal{M}H_*$	Morse homology
<code>\ChCW</code>	${}^{CW}C_*$	CW chain complex
<code>\bCW</code>	${}^{CW}\partial$	CW boundary
<code>\HCW</code>	$H^{CW}H_*$	CW homology
<code>\conn</code>	δ	connecting homom. (δ)
<code>\K</code>	K	K-theory
<code>\KO</code>	KO	real K-theory
<code>\KU</code>	KU	complex K-theory
<code>\Epq{r}{p,q}</code>	$E_{p,q}^r$	SS entry
<code>\drpq{r}</code>	d^r	SS differential
<code>\Crit</code>	Crit	critical point set
<code>\ind</code>	ind	Morse index
<code>\grad</code>	grad	gradient
<code>\euler</code>	e	Euler number (roman e)
<code>\ii</code>	i	imaginary unit (roman i)

1 Vector Bundles and Riemannian Geometry

Due to differential geometry and the theory of vector bundles being areas of mathematics used by pure mathematicians interested in analytical aspects like curvature, by topologists interested in invariants like Euler characteristics, physicists using the spaces as playgrounds for general relativity theories, and many others, there are numerous competing notational conventions. This thesis lies at the tripoint of differential geometry, dynamical systems, and algebraic topology. Hence, there is no a priori preferred notational convention. In this chapter, we establish the necessary background in the theory of vector bundles and Riemannian manifolds whilst not proving all the basics; we establish the notation used throughout this thesis. We establish manifolds and vector bundles in parallel, as they are interdependent on one another.

1.1 Differentiable Structures on Topological Manifolds

We begin by defining a topological manifold. Afterwards, we equip that space with with increasingly more structure and hence reduce the generality until we arrive at the playground of the thesis:

Definition 1.1.1 (Topological Manifold). A second countable Hausdorff space M is called a **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, for all $p \in M$ there exists an open neighbourhood $U \subseteq M$ of p , an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : U \rightarrow V$ that is a homeomorphism. We call the map $\varphi : U \rightarrow V$ a **chart around p on M** , and φ^{-1} a **local coordinate system around p on M** .

Definition 1.1.2 (Differentiable Manifold). Let M be a topological manifold of dimension m .

1. A **differentiable atlas of class $r \in \mathbb{N} \cup \{\infty\}$** is a family of charts $\mathfrak{A} = (\varphi_i : U_i \rightarrow V_i)_{i \in I}$ such that
 - a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M .
 - b) For every pair $(i, j) \in I^2$, the **transition function**:

$$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\rightarrow \varphi_i(U_i \cap U_j) \\ x &\mapsto (\varphi_i \circ \varphi_j^{-1})(x) \end{aligned}$$

is differentiable of class r .

We call such an atlas a C^r -atlas.

2. Two C^r -atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r -atlas.

A **differentiable structure of class r** on M is an equivalence class c of C^r -atlases. For $r = \infty$, we call the pair (M, c) a smooth manifold.

Corollary 1.1.3. Every transition function φ_{ij} , $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not merely a homeomorphism but also a diffeomorphism. This follows from the fact that $\varphi_{ij}^{-1} = \varphi_{ji}$ which provides smooth inverses.

Definition 1.1.4 (Smooth Functions). Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuous function

$$f : U \rightarrow \mathbb{R}$$

differentiable of class r , if for any chart (and hence for all) $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r . For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f : U \rightarrow \mathbb{R} \text{ continuous} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 1.1.5. Let (M, c) be a smooth manifold of dimension m and $U \subseteq M$ be an open subset. With pointwise defined operations, the set $(\mathcal{E}(U), +, \cdot, \circ)$ becomes an $\mathbb{R}$ -algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$ -algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \text{res}_V^U : \mathcal{E}(U) &\rightarrow \mathcal{E}(V) \\ f &\mapsto f|_V. \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. $\square$

Definition 1.1.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in p** , if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_W = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in p** . We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{ open}, p \in U} \mathcal{E}(U) \right) / \sim_p$$

for the set of germs, and call it the **stalk in p** . Where $\sum$ denotes the co-product (disjoint union) in **Top**.

Corollary 1.1.7. For a smooth manifold (M, c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$ -algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \text{eval}_p : \mathcal{E}_p(M) &\rightarrow \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\text{eval}_p)$. Hence, the pair $(M, \mathcal{E})$ gives us a locally ringed space.

Proof. We only need to verify the local ring structure of the stalks. An element $f_p \in \mathcal{E}_p(M)$ is invertible if and only if $f(p) \neq 0$, equivalently if $f_p \notin \ker(\text{eval}_p)$. Thus $\mathcal{E}_p(M)/\ker(\text{eval}_p)$ is a field, which proves that $\ker(\text{eval}_p)$ is maximal. $\square$

Definition 1.1.8. Let (M, c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$ -linear map $\delta : \mathcal{E}_p(M) \rightarrow \mathbb{R}$ a **derivation**, if it satisfies the Leibniz rule:

$$\delta(f_p \cdot g_p) = \delta(f_p)g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f_p, g_p \in \mathcal{E}_p(M).$$

We call $\text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$ the set of derivations and give it an $\mathbb{R}$ -vector space structure by pointwise operations. We define the **tangent space of M at p** to be the vector space

$$T_p M := \text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R}).$$

Corollary 1.1.9. Let (M, c) be a smooth manifold of dimension m and $\varphi : U \rightarrow V$ be a chart around p with $x_0 = \varphi(p)$ (where $\varphi \in \mathfrak{A} \in c$). Then we define the derivation

$$\begin{aligned} \partial_i(p) : \mathcal{E}_p(M) &\rightarrow \mathbb{R} \\ f_p &\mapsto \partial_i(p)(f_p) := \frac{\partial(f \circ \varphi^{-1})}{\partial x_i} \Big|_{x_0}. \end{aligned}$$

This is well-defined and hence gives a tangent vector. Moreover, the family

$$(\partial_1(p), \dots, \partial_m(p))$$

defines a basis of $T_p M$. Hence, the dimension of $T_p M$ is m .

This is a good point to take a break from smooth manifolds and introduce the structure of vector bundles.

Analogously to the definition of a smooth manifold, we start by defining families of vector spaces and equip them with increasingly more structure:

Definition 1.1.10 (Family of Vector Spaces). Let X be a topological space. A **family of vector spaces over X** is a topological space E equipped with :

- (i) a continuous function $\pi : E \rightarrow X$,
- (ii) a finite dimensional vector space structure on each fiber $E_x := \pi^{-1}(x)$ that is compatible with the subspace topology on $E_x \subseteq E$.

We call the map π the projection, E the total space and X the base space.

Definition 1.1.11 (Sections). Let $\pi : E \rightarrow X$ be a family of vector spaces. A function $f : X \rightarrow E$ such that for all $x \in X$ we have $\pi \circ f(x) = x$ is called a **section of E** . We denote the space of all sections by $\Gamma(E)$.

Remark 1.1.12. Let $\pi : E \rightarrow X$ be a family of vector spaces. By locality of the definition, for all open $U \subseteq X$ the restriction $\pi|_U : \pi^{-1}(U) \rightarrow U$ defines a new family of vector spaces. With this we can also have sections of the restricted bundle and the collection of those is denoted by $\Gamma(U)$.

Definition 1.1.13 (Pullback of a Family of Vector Spaces). Let $\pi : E \rightarrow Y$ be a family of vector spaces, X a topological space and $f : X \rightarrow Y$ a continuous function. Then we define the pullback of the family $f^*\pi : f^*(E) \rightarrow X$ as follows: The total space $f^*(E)$ is defined as the subspace of $(x, e) \in X \times E$ such that $f(x) = \pi(e)$, and the map $f^*\pi$ is the projection to the second factor restricted to said subspace.

Corollary 1.1.14. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be continuous. Then there is a natural isomorphism

$$g^*f^*(E) \cong (g \circ f)^*(E).$$

Definition 1.1.15 (Trivial Bundle). If a family $\pi : E \rightarrow X$ is isomorphic to a product family $\tilde{\pi} : X \times V \rightarrow X$ we call it a **trivial family**.

Definition 1.1.16 (Vector Bundle). If for every $x \in X$ there is a neighborhood U containing x such that $E|_U := i^*(E)$, where $i : U \hookrightarrow X$ is the inclusion, is trivial, we call the family a **vector bundle**. Moreover, for any continuous map $f : Y \rightarrow X$, the pullback $f^*(E)$ is again a vector bundle.

Remark 1.1.17. Let $\pi : E \rightarrow X$ be a vector bundle. Then the space of sections $\Gamma(E)$ is a (in general, infinite-dimensional) $\mathbb{R}$ -vector space. Furthermore, and more relevant to the purposes of this thesis, we view $\Gamma(E)$ (and $\Gamma(U)$ for open subsets) as a $C(X)$ (or $C(U)$) module. The latter denotes the ring of continuous functions from E (or U) to $\mathbb{R}$.

Example 1.1.18. For any smooth manifold M of dimension m , we can equip the union

$$TM := \bigsqcup_{p \in M} T_p M$$

with a topology, such that $\pi : TM \rightarrow M$ defines a vector bundle which we call **tangent bundle**. For this we first define the trivialisations and then equip TM with a topology to make them homeomorphisms: Let $\varphi : U \rightarrow V$ be a chart on M . If we denote the i -th coordinate function of φ by φ_i we can define the bijection

$$\tilde{\varphi} : \pi^{-1}(U) \rightarrow V \times \mathbb{R}^m, \quad \xi_p \mapsto \left(\varphi(p), \begin{pmatrix} \xi_p(\varphi_1) \\ \vdots \\ \xi_p(\varphi_m) \end{pmatrix} \right).$$

Now we declare a subset $W \subseteq TM$ to be open, if and only if for all charts φ the set $\tilde{\varphi}(W \cap \pi^{-1}(U)) \subset V \times \mathbb{R}^m$ is open. We can now extend and give meaning to the notation of the local basis vectors $\partial_i(p) \in T_p M$: For any chart $\varphi : U \rightarrow V$ we have the local sections

$$\partial_i : U \rightarrow TM, \quad p \mapsto \partial_i(p).$$

Furthermore, TM is a $2m$ -dimensional manifold. Using a smooth atlas on M yields a smooth atlas on TM making it a smooth manifold. Notice that the family $(\partial_i)_i$ is a basis of the $C(U)$ -module $\Gamma(U)$. This follows by expressing any section in local coordinates via $\tilde{\varphi}$.

Notice that TM as a bundle satisfies more than just the bundle definition: Taking two different charts induces two different local trivialisations. Transitioning between them is not just fibrewise linear, but also smooth. This is captured in the definition:

Henceforth we suppress the differentiable structure from the notation.

Definition 1.1.19 (The Derivative). Let M and M' be smooth manifolds. We call a continuous function $f : M \rightarrow M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in \mathcal{C}$ and $\varphi' \in \mathfrak{A}' \in \mathcal{C}'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \rightarrow V'$$

are smooth. A given smooth function induces a smooth map between the tangent bundles as manifolds as follows:

$$df : TM \rightarrow TM', \quad df_p(\xi_p)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi_p \in T_p M$, $g_p \in \mathcal{E}_p(M')$. The function df is called the **derivative** or **pushforward** of f .

Definition 1.1.20 (Smooth Vector Bundles). A **smooth vector bundle of rank r** is a surjection $\pi : E \rightarrow M$ where the codomain (or base space) is a smooth manifold together with a family of continuous functions (or local trivialisations)

$$\Phi_i : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^r$$

such that $\bigcup_i U_i = M$ and for all overlapping $U_i \cap U_j \neq \emptyset$, the transition functions

$$\Phi_j \circ \Phi_i^{-1} : \pi^{-1}(U_i \cap U_j) \rightarrow (U_i \cap U_j) \times \mathbb{R}^r$$

are of the form

$$\Phi_j \circ \Phi_i^{-1}(p, v) = (p, g_{i,j}(v)), \quad \text{where} \quad g_{i,j} : (U_i \cap U_j) \rightarrow \text{GL}(r, \mathbb{R})$$

is smooth.

Corollary 1.1.21. A smooth vector bundle is a smooth manifold and also a vector bundle. The latter is ensured by the codomain of $g_{i,j}$ which is also called the **structure group**. We can enforce structure on the bundle by reducing the structure group. As a preview: reducing it to $\text{GL}^+(k, \mathbb{R})$ the linear maps with positive determinant means that we can induce the concept of orientations. Reducing it to $\text{O}(k)$ lets us equip it with a Riemannian metric.

Definition 1.1.22. Let $\pi : E \rightarrow M$ be a smooth vector bundle of rank r and let

$$\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$$

be a local trivialisation. Then a section is called smooth if it is a smooth map of manifolds. A family of smooth local sections $(s_1, \dots, s_r)$ with $s_i \in \Gamma(U)$ is called a **local frame** if they form a basis of the $C(U)$ -module $\Gamma(U)$. Such a family is obtained from Φ by defining $s_i(p) := \Phi^{-1}(p, e_i)$. Similar, from a local frame we can obtain a local trivialisations by defining:

$$\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r, \quad v = \sum_{i=1}^r \lambda_i s_i(\pi(v)) \mapsto (\pi(v), \lambda_1, \dots, \lambda_r).$$

Example 1.1.23. The local trivialisations ∂_i of the tangent bundle of a smooth manifold M form a local frame and make the tangent bundle into a smooth vector bundle.

Definition 1.1.24 (Compact-open topology). Let X, Y be topological spaces. Then the set of continuous maps $C(X, Y)$ can be equipped with the topology that has

$$\{V(K, U) \mid K \subseteq X \text{ compact}, U \subseteq Y \text{ open}\} \text{ where } V(K, U) := \{f \in C(X, Y) \mid f(K) \subseteq U\}$$

as a subbasis. This coincides with the topology of $\text{Hom}(V, W)$ as a subset of $\mathbb{C}^n$ after identification with the corresponding matrix.

Definition 1.1.25 (Bundle Homomorphisms). Let $\pi_1 : E_1 \rightarrow X$ and $\pi_2 : E_2 \rightarrow X$ be two vector bundles over the same base space X . A **morphism of vector bundles over the same base space** is a continuous map $f : E_1 \rightarrow E_2$ such that

1. $\pi_2 \circ f = \pi_1$, and hence for each $x \in X$ the restriction to the fibers $f_x : (E_1)_x \rightarrow (E_2)_x$ is well defined.
2. For each $x \in X$, f_x is a vector space homomorphism.

Definition 1.1.26 (Homomorphism Bundle). Let W, V be vector spaces and $E = V \times X$, $F = W \times X$ the product bundles over X . Now any bundle homomorphism $\varphi : E \rightarrow F$ determines a map $\Phi : X \rightarrow \text{hom}(V, W)$ by sending $x \mapsto \varphi_x$. With respect to the compact-open topology, this Φ is continuous. Furthermore, a continuous map $\Phi : X \rightarrow \text{hom}(V, W)$ determines a bundle homomorphism $\varphi : E \rightarrow F$.

Proof. Let $\{e_i\}$ and $\{f_i\}$ be bases of V and W . Then each $\Phi(x)$ is a matrix $\Phi_{i,j}(x)$, such that:

$$\Phi(x)e_i = \sum_j \Phi(x)_{i,j} f_j.$$

Now Φ is continuous if and only if $\Phi_{i,j}$ is continuous. Since $\varphi(x, v) = (x, \Phi(x)v)$ this is equivalent to φ being continuous. $\square$

Lemma 1.1.27 (Isomorphic Vector Bundles). *Let E and F be vector bundles and $\varphi : E \rightarrow F$ be a homomorphism. Then φ is an isomorphism if and only if all φ_x are isomorphisms.*

Proof. To show that it is an isomorphism we need a continuous inverse, equivalently, we need to show that φ is a homeomorphism. First of all we can deduce, that φ is bijective: Assume it wasn't injective, then there would be a fibre in which φ_x is not injective. Assume that it wasn't surjective. Then there would be a $w \in F$ that has no preimage which means that in the fiber containing w there is no isomorphism. Thus, there is a unique inverse and it remains to verify that it is continuous, which is a local condition. Therefore we can assume that E and F are product bundles. Then we have a continuous map $\Phi : X \rightarrow \text{Hom}(V, W)$. Furthermore, since all φ_x are isomorphisms, we can restrict

the image to $\text{Iso}(V, W)$ which is an open subset of $\text{Hom}(V, W)$. Now we can define the map $x \mapsto \Phi(x)^{-1}$ which is continuous since taking inverses is continuous. By the corollary above, we obtain a continuous map $\psi : F \rightarrow E$ by sending $(x, w) \mapsto (x, \Phi(x)^{-1}(w))$, which is the inverse. $\square$

Example 1.1.28. Let M be a smooth manifold of dimension m , $\varphi : U \rightarrow V$ a chart and $\tilde{\varphi} : TU \rightarrow V \times \mathbb{R}^m$ be the corresponding bundle chart. Then it follows from the lemma, that TU is isomorphic to $V \times \mathbb{R}^m$. This is verified by observing that $\partial_i(p)$ is a basis of $T_p U$ for all $p \in U$. In this case $\tilde{\varphi}(\partial_i) = e_i$ giving us isomorphisms in each fiber.

Now we want to construct new bundles from old ones. There are many operations to construct new vector spaces such as the tensor product, direct sum, Hom-Spaces and many more. Instead of individually proving that those operations applied fiberwise yield new bundles, we subsume these operations within the following definition:

Definition 1.1.29 (Continuous Functor). Let T be a covariant endofunctor in the category of finite-dimensional vector spaces. We call T a **continuous functor**, if for all V, W the map $T : \text{hom}(V, W) \rightarrow \text{hom}(T(V), T(W))$ is continuous. All vector spaces are given the topology induced by their isomorphism to real Euclidean space. If E is a vector bundle we define the set:

$$T(E) := \bigcup_{x \in X} T(E_x).$$

Now we want to equip this with a topology such that $T(E)$ is a bundle over the same base space as E .

Definition 1.1.30 (Constructing new bundles). We start by assuming that $E = X \times V$. In that case we define the topology on $T(E)$ as the product topology $X \times T(V)$. Now suppose that $F = X \times W$ is another product bundle and $f : E \rightarrow F$ is a homomorphism. Let $\Phi : X \rightarrow \text{hom}(V, W)$ be the corresponding map. Then $T\Phi : X \rightarrow \text{hom}(T(V), T(W))$ is continuous by assumption. Thereby, $T(\varphi) : X \times T(V) \rightarrow X \times T(W)$ is continuous. If φ is an isomorphism, then $T(\varphi)$ is also an isomorphism. That follows because it is continuous and an isomorphism fibrewise.

Now assume that E is trivial. Then we can choose an isomorphism $\alpha : E \rightarrow X \times V$ and induce a topology on $T(E)$ by pulling it back via $T(\alpha) : T(E) \rightarrow X \times T(V)$. This topology does not depend on α by the above. For general vector bundles we do this construction locally.

If $f : X \rightarrow Y$ is continuous, there is a natural isomorphism

$$f^*T(E) \cong Tf^*(E)$$

This construction is carried out in detail by Atiyah in [Ati89, Chapter 1.2] and can be generalised for continuous functors with several variables both co- and contravariant. Hence this procedure gives us for vector bundles E and F over the same base space the new bundles:

- (i) $E \oplus F$, their direct sum
- (ii) $E \otimes F$, their tensor product
- (iii) $\text{Hom}(E, F)$
- (iv) E^* , the dual bundle of E
- (v) $\Lambda^i(E)$ the i -th exterior power

and by the natural isomorphism we have natural isomorphisms:

- (i) $E \oplus F \cong F \oplus E$
- (ii) $E \otimes F \cong F \otimes E$
- (iii) $E \otimes (F \oplus F') \cong E \otimes F \oplus E \otimes F'$
- (iv) $\text{Hom}(E, F) \cong E^* \otimes F$

Corollary 1.1.31. Let $\pi_E : E \rightarrow X$ and $\pi_F : F \rightarrow X$ be two vector bundles over X . Then, the space $\Gamma(\text{Hom}(E, F))$ can be identified with the space of bundle morphisms $\text{hom}(E, F)$ since for any section $s : X \rightarrow \text{Hom}(E, F)$ we can define the bundle homomorphism $\tilde{s} : E \rightarrow F$ by sending $v \mapsto s_{\pi_E(v)}(v)$. This is a homomorphism since $\pi_F(\tilde{s}(v)) = \pi_E(v)$.

Example 1.1.32 (Cotangent bundle). Let M be a smooth manifold of dimension m . Then we can define the **cotangent bundle** T^*M to be the dual of the tangent bundle. A local frame of said bundle is given by the fiberwise dual of the frame (∂_i) . Sections into the tangent bundle are called **vector fields** and sections into the cotangent bundle are called **one-forms**.

Definition 1.1.33. Let V be a linear manifold (i.e. a finite-dimensional vector space). Then for any $v \in V$ we have the canonical maps

$$R_v : V \rightarrow T_v V$$

$$x \mapsto \left(f_v \mapsto \frac{\partial}{\partial t} \Big|_{t=0} f(v + tx) \right)$$

Let $L : V \rightarrow V$ be a linear map viewed as a smooth map between manifolds. Then we have the identity:

$$dL_v \circ R_v = R_{L(v)} \circ L.$$

This is proved by a short calculation:

$$\left(dL_v(R_v(x)) \right) (f_{L(v)}) = R_v(x) (f \circ L) = \frac{\partial}{\partial t} \Big|_{t=0} f \circ L(v + tx) = \underbrace{\frac{\partial}{\partial t} \Big|_{t=0} f(L(v) + tL(x))}_{=R_{L(v)} \circ L(x)}$$

Now let $v, v' \in V$. Then we have the translation

$$tr_{v,v'} : V \rightarrow V; \quad x \mapsto (x + (v' - v))$$

This allows us to relate R_v and $R_{v'}$ as follows:

$$(dtr_{v,v'}|_v(R_v(x)))(f_{v'}) = \frac{\partial}{\partial t} \Big|_{t=0} (f \circ tr_{v,v'}(v + tx)) = \frac{\partial}{\partial t} \Big|_{t=0} (f(v' + tx)) = R_{v'}(x)(f_{v'}).$$

Hence:

$$R_{v'} = dtr_{v,v'} \circ R_v.$$

The **linear identification** is the map Q given on the whole tangent bundle:

$$Q : TV \rightarrow V; \quad \xi_p \mapsto R_p^{-1}(\xi_p),$$

and satisfies for any linear $L : v \rightarrow V$

$$Q \circ dL = L \circ Q.$$

Corollary 1.1.34. Let $f : M \rightarrow \mathbb{R}$ be smooth. Then we can interpret df as a one-form. To be precise, let $Q : T\mathbb{R} \rightarrow \mathbb{R}$ denote the linear identification with respect to the identity on $\mathbb{R}$. For $v \in \Gamma(TM)$ we have

$$(Q \circ df)(v) = v(f)$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We retain this notation so that vector fields can take functions as an input.

Proof. Let $\varphi : U \rightarrow V$ be a chart of M . We prove this by showing it for the local sections ∂_i and the general case follows by $C(U)$ -linearity, since those sections form a local frame. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$df_p(\partial_i(p))(g_p) = \partial_i((g \circ f)_p) = \frac{\partial}{\partial x_i} \Big|_{x_0} (g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x_{f_p(p)}} (g_p) \cdot \partial_i(p)(f_p)$$

Hence, applying Q to the calculation yields $(Q \circ df)(v) = v(f)$. $\square$

Definition 1.1.35 (A Metric). Let M be a smooth manifold. A section $g \in \Gamma(T^*M \otimes T^*M)$ into the tensor product of the dual tangent bundle with itself is a **Riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non-degenerate**, meaning if for a $v_p \in T_pM$ we have $g_p(v_p, w_p) = 0$ for all $w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ for all $v_p, w_p \in T_pM$.
3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0$ for all v_p with equality only for $v_p = 0$.

We call a tuple (M, g) a **Riemannian manifold**.

Remark 1.1.36. Equipping a smooth manifold M of dimension m with a Riemannian metric is equivalent to reducing the structure group of the tangent bundle from $GL(m)$ to $O(m)$. To be precise: A Riemannian metric g lets us canonically choose an atlas with structure group $O(m)$ and an atlas with such a structure group lets us canonically choose a Riemannian metric.

Proof. We start by assuming that we have an atlas with structure group $O(m)$. In Definition 1.1.22 we saw that a covering of local trivializations is equivalent to a covering of smooth frames. Now in a single trivialization

$$\Phi : \pi^{-1}(U) = TU \rightarrow U \times \mathbb{R}^m$$

we call the projection to the $\mathbb{R}^m$ -part (the second factor) π_2 and define

$$\begin{aligned} g : U &\rightarrow T^*U \otimes T^*U \\ p &\mapsto g_p^\Phi : T_pU \times T_pU \rightarrow \mathbb{R} \\ &\quad (\nu_p, \xi_p) \mapsto g_p^\Phi(\nu_p, \xi_p) := \pi_2(\Phi(\nu_p)) \cdot \pi_2(\Phi(\xi_p)), \end{aligned}$$

where $\cdot$ denotes the standard scalar product. This gives us locally a Riemannian metric. If we do this for all trivializations we have to make sure, that the result is well defined. So assume that two trivializations Φ and Ψ overlap in V . The structure group is $O(m)$, $\Phi = R\Psi$ for some function

$$R : V \rightarrow O(m).$$

Let p be in said overlapping and (ν_p, ξ_p) be from $T_pM \times T_pM$. Then since R is orthogonal we can calculate the independence from the trivialization:

$$g_p^\Phi(\nu_p, \xi_p) = \pi_2(R\Psi(\nu_p)) \cdot \pi_2(R\Psi(\xi_p)) = \pi_2(\Psi(\nu_p)) \cdot \pi_2(\Psi(\xi_p)) = g_p^\Psi(\nu_p, \xi_p).$$

To get an orthonormal atlas from a smooth one given a Riemannian metric, we start by building a covering of local frames from the trivialisations. We then apply Gram-Schmidt to the local frames to get orthonormal frames. Now a covering of orthonormal frames gives a covering of trivializations, where the structure group consists of basis changes between orthonormal bases and hence is $O(m)$. $\square$

Definition 1.1.37 (The Gradient). Assume that (M, g) is a smooth manifold together with a Riemannian metric. Let $f : M \rightarrow \mathbb{R}$ be a smooth function and $Q : T\mathbb{R} \rightarrow \mathbb{R}$ the linear identification. We define its **gradient** to be a vector field $\text{grad}(f) \in \Gamma(TM)$ such that for any vector field V :

$$Q \circ df(V) = g(\text{grad}(f), V).$$

Notice how the non-degeneracy of g ensures existence and uniqueness.

Definition 1.1.38. Let (M, g) be a Riemannian manifold and $\varphi : U \rightarrow V$ be a chart. We define the smooth functions $g_{ij} : U \rightarrow \mathbb{R}$ to be:

$$g_{ij}(p) = g_p(\partial_i(p), \partial_j(p)).$$

Then if two vector fields v, w over U are given with $v = \sum v_i \partial_i$ and $w = \sum w_i \partial_i$ we can calculate the metric locally:

$$g(v, w) = \sum_{i,j} g_{ij} v_i w_j : U \rightarrow \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure as can be seen by the construction via Cramer's rule) to get smooth functions

$$g^{ij} : U \rightarrow \mathbb{R}$$

$$p \mapsto g^{ij}(p) = \left((g_{kl}(p))_{kl}^{-1} \right)_{ij}$$

Remark 1.1.39. Due to the rare appearance of big summations with vectors and co-vectors in this work, we omit the Einstein summation convention and assign all coordinate entries lower indices.

Lemma 1.1.40 (Local Description of the Gradient). *Let (M, g) be a smooth Riemannian manifold of dimension m and $\varphi : U \rightarrow V$ be a chart. Then the gradient has the local form:*

$$\text{grad}(f) = \sum_{i,j} g^{ij} (\partial_i(f)) \cdot \partial_j.$$

Proof. Assume that $v, w \in \Gamma(TU)$ such that $v = \sum v_i \partial_i$ and $w = \sum w_i \partial_i$ (Here v_i and w_i are smooth functions $U \rightarrow \mathbb{R}$). Then $g(v, w) = \sum_{i,j} g_{ij} v_i w_j$. Suppose that $\text{grad}(f) = \sum_j G^j \partial_j$ and Q is the linear identification $T\mathbb{R} \rightarrow \mathbb{R}$ in each tangent space. Then by definition of the gradient and corollary 1.1.34 we have:

$$\partial_j(f) = Q \circ df(\partial_j) = g(\text{grad}(f), \partial_j) = g\left(\sum_i G^i \partial_i, \partial_j\right) = \sum_i g_{ij} G^i$$

Reading this as a pointwise matrix product:

$$(G^1, \dots, G^m)(g_{ij}) = (\partial_1(f), \dots, \partial_m(f))$$

which gives $(G^1, \dots, G^m) = (\partial_1(f), \dots, \partial_m(f))(g^{ij})$.

But then we can conclude that $G^j = \sum_i g^{ij} \partial_i(f)$. □

Definition 1.1.41 (A Dynamical System). Let M be a smooth manifold. A **dynamical system on M** (sometimes also called a **flow on M**) is given by a smooth map $\varphi : \Omega \rightarrow M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cap \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t, p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t, p))) = \varphi(s + t, p)$$

3. For all p we require $\varphi(0, p) = p$.

We will write $\varphi_t(p) := \varphi(t, p)$ to keep the notation manageable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 1.1.42. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \rightarrow \text{Diff}(M), \quad t \mapsto \varphi_t$$

is a homomorphism. To verify well-definedness notice how for all $t \in \mathbb{R}$, φ_{-t} is an inverse of φ_t . The map ρ is called a **one-parameter group of diffeomorphisms**.

Definition 1.1.43 (Vector Field of a Dynamical System). Let $\varphi : \Omega \rightarrow M$ be a dynamical system on a smooth manifold M . For a chart $\psi : U \rightarrow V$ of M we have the chart $\text{id} \times \psi$ of Ω . For this chart we denote the associated local frame by $(\partial_t, \partial_1, \dots, \partial_m)$. Using this we define the mapping

$$\begin{aligned} M &\rightarrow TM \\ p &\mapsto X(p) := d\varphi(\partial_t(0, p)) \end{aligned}$$

and call it the **associated vector field**. This is in fact a smooth vector field.

Definition 1.1.44 (Dynamical System of a Vector Field). Let $X \in \Gamma(TM)$ be a global vector field on the smooth manifold M . Then we call a dynamical system φ on M the **associated dynamical system** to X , if for all $t \in \mathbb{R}$ and $x \in M$

$$d\varphi(\partial_t(t, x)) = X(\varphi_t(x)).$$

Remark 1.1.45. Regarding the left-hand side of the definition $d\varphi(\partial_t(t, x))$ we could also differentiate the map

$$t \mapsto \varphi_t(x),$$

for fixed x at t and evaluate this at the constant vector field $t \mapsto R_t(1) \in T_t\mathbb{R}$ from definition 1.1.33.

These two constructions are mutually inverse, giving the following theorem:

Theorem 1.1.46. *Let M be a smooth manifold. Then there is a one-to-one correspondence between global dynamical systems and vector fields given by the constructions above.*

2 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [BH04]. In Morse theory, we will always deal with a smooth function $f : M \rightarrow \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 2.0.1 (Critical Points). A point $p \in M$ at which the smooth map $df_p = 0$ is called **critical**. By $\text{Crit}(f)$ we denote the **set of critical points**.

Definition 2.0.2 (The Hessian). Let $f : M \rightarrow \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \text{Crit}(f)$ is a symmetric bilinear function

$$\text{Hess}_p(f) : T_p M \times T_p M \rightarrow \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_p M$ be two tangent vectors. Now choose two vector fields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. We then define the Hessian to be:

$$\text{Hess}_p(f)(\xi_p, \zeta_p) = \Xi(Z(f))(p)$$

See 1.1.34 for the notation of applying a function to a vector field. In corollary ?? we will give a local description telling us that this definition is well defined, meaning independent of the Extensions Ξ and Z .

Definition 2.0.3. A smooth function $f : M \rightarrow \mathbb{R}$ is called **Morse**, if for any critical point, the Hessian is a symmetric bilinear and non-degenerate map.

For such a map there exists the following theorem:

Theorem 2.0.4 (Sylvesters Law of Inertia). *Let $A \in \text{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \text{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\text{rank}(A) = k + l$.

References

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